Students's Name: ……

Signature: ……Class & Stream: ……..……..

P510/1
PHYSICS
Paper 1
(Theory)
2 hours 15 minutes

UGANDA STEM TEACHERS NETWORK (USTEN)

Uganda Advanced Certificate of Education

PHYSICS REVISION ASSESSMENT SET ONE

Paper 1

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

- This examination paper consists of **THREE compulsory** items.
- Responses to **each** item should start on a **fresh page**.
- Silent non-programmable scientific calculators may be used.

ITEM 1

A large engineering company in Uganda intends to establish an industrial radiography facility for inspecting welded joints in steel pipelines. The proposed facility plans to use a radioactive source of caesium-137, $^{137}_{55}Cs$, to produce penetrating radiation for detecting hidden defects in the welds.

The isotope undergoes beta decay to form barium-137. The radioactive source also produces gamma radiation during the decay process. Before the facility is approved, the radiation-safety committee requires an assessment of the proposed source.

The committee wants to determine whether the proposed quantity of the isotope satisfies the activity requirements and whether the energy released during a decay is adequate for the intended application.

The company also intends to install an X-ray radiography unit as an alternative method of inspecting some components. An Advanced Level Physics learner has been invited to explain the operation of this device.

Support material

- Mass of $^{137}_{55}Cs$ = 136.90709 U
- Mass of $^{137}_{56}Ba$ = 136.90583 U
- Half-life of $^{137}_{55}Cs$ = 30.05 years
- Mass of source = 20 g
- Molar mass of ^{137}Cs = 137 g mol^{-1}
- Avogadro constant = 6.02×10^{23} mol^{-1}
- 1 U = 931 MeV/c^2
- Maximum permitted activity = 5.8×10^{13} Bq
- Minimum useful energy per decay = 1.0 MeV

TASK:

You have been selected to assist the radiation-safety committee.

a) Analyse the given information to determine whether the proposed caesium-137 source satisfies the specified requirements.

b) The company proposes to use an X-ray radiography unit for inspecting selected components. Explain the structure and working principle of an X-ray tube, showing how electrons are used to produce X-rays suitable for imaging internal structures.

ITEM 2

Uganda is expanding its use of satellite technology for monitoring agricultural land, forests and water resources. A new earth-observation satellite of mass 300 kg is to be placed in a circular parking orbit above a launch station in central Uganda. The project engineers need to establish the orbital height corresponding to a specified mechanical energy before the satellite is released.

For safety purposes, the satellite is required to have a total mechanical energy of -5.0×10^9 J in its parking orbit. The engineers also need to understand how changes in the mechanical energy of the satellite would affect its orbital position.

A field engineer at the launch station has been instructed to deliver some final technical documents to the control centre within 5 minutes. He drives a service

vehicle of mass 1800 kg along a straight access road. The engine develops a driving force of 900 N and has a power rating of 18 kW. The vehicle starts from rest and accelerates uniformly until it reaches the speed corresponding to the engine's stated power and driving force. It covers the first 1.0 km while accelerating and then the driver applies the brakes over the remaining 0.60 km.
The coefficient of friction between the tyres and the road is 0.030.
The project supervisor has also requested the engineer to suggest a simple locally manufactured aerial-survey device capable of carrying a camera and producing an upward lift force.

Support Material

- Mass of satellite = 300 kg
- Required satellite mechanical energy = -5.0×10^9 J
- Radius of Earth, RE = 6.4×10^6 m
- Acceleration due to gravity, g = 9.81 m s^{-2}
- Coefficient of friction = 0.030
- Mass of vehicle = 1800 kg
- Engine power = 18 kW
- Driving force = 900 N
- Acceleration distance = 1.0 km
- Braking distance = 0.60 km

TASK

You have been selected to assist the project team.

a) Analyse the information provided and determine the necessary information required by the project manager.
b) Explain how a simple propeller-driven aerial-survey drone could be constructed using locally available materials and describe how it produces the upward lift force required to carry the camera.

ITEM 3

A secondary school recently renovated its library by installing large double-glazed windows to improve natural lighting and ventilation. During sunny afternoons, learners complained that the library became uncomfortably hot. The contractor maintained that the new windows provided adequate thermal insulation and that the temperature inside the library would remain below 22°C when the outside surface of the glass reached 34°C.
The windows consist of two glass panes, each 5 mm thick, separated by a 10 mm layer of dry air. The total window area is 24 m^2. Measurements showed that the windows receive solar radiation of intensity 900 W m^{-2}, and approximately 45% of the incident solar radiation is absorbed by the window system.
The contractor stated that the thermal conductivities of glass and dry air were 0.78 W m^{-1} K^{-1} and 0.026 W m^{-1} K^{-1}, respectively.
The Physics teacher questioned whether the stated thermal conductivity of the glass and the claimed indoor temperature were realistic. A Senior Six learner was therefore asked to investigate the claims using a sample of the same glass material. For the investigation, the learner used a glass sample of cross-sectional area 0.040 m^2 and thickness 20 mm. When a steady temperature difference of 15 K was

maintained across the sample, the rate of heat transfer through it was measured as 24 W.

The school also intends to reduce solar heating by modifying the outer surface of the windows. In addition, the administration wants an electronic device near the library entrance to detect unauthorized entry.

Support Material

- Thermal conductivity of dry air = 0.026 W m^{-1} K^{-1}
- Claimed thermal conductivity of glass = 0.78 W m^{-1} K^{-1}
- Solar radiation intensity = 900 W m^{-2}
- Fraction of solar radiation absorbed = 45%
- Window area = 24 m^2

TASK

You have been selected by the Physics teacher.

a) Using the data provided, analyse the situation and determine whether the contractor's claims concerning the thermal performance of the windows are justified. Your analysis should include determining the thermal conductivity of the glass sample and considering the heat transfer through the double-glazed window.

b) Explain how the outer surface of the window could be modified to reduce the amount of solar energy absorbed and describe the operation of an infrared motion detector that could be installed near the library entrance to detect unauthorized entry.

END

KATO IVAN WUUNA
+256 788 463 703
DOWNLOAD MORE RESOURCES FROM WUNNA DIGITAL LIBRARY